ARES-X 0.9.16

Windows Distribution — User Guide

N2MDX | X-TRACKER • WSJT-X Integration • VAC Spectrum / Waterfall

Read this guide before connecting a radio or using any reply/transmit control.

1. What is included

ARES-X is an experimental Windows amateur-radio digital-mode and station-tools application by N2MDX. The ZIP includes Python source, assets, dependencies, a Windows build script, and these instructions.

This is a **source/build distribution**, not a prebuilt installer or precompiled EXE. Build it on the Windows PC using the included script.

2. Build and launch on Windows 10/11

Requirements: 64-bit Windows 10/11, 64-bit Python 3.11 recommended, and internet access for the first dependency installation.

1. Extract the entire ZIP to a normal folder, such as C:\\Radio\\ARES-X-0.9.16\\. Do not run from the ZIP preview.
2. Install 64-bit Python 3.11 and enable the Python launcher if offered.
3. Open the extracted ARES-X-Windows-0.9.3 folder.
4. Double-click **build_windows.bat** and wait for it to finish.
5. Launch **dist\\ARES-X\\ARES-X.exe**. Keep the entire dist\\ARES-X folder together.

Optional source launch from Command Prompt in the project folder:

```
py -3.11 -m pip install -r requirements.txt
py -3.11 ares_x_windows.py
```

3. Radio CAT setup

Configure the radio's CAT or virtual serial interface first. Close other programs that are using the same COM port. In ARES-X select the matching radio profile, COM port, and baud rate, then click **CONNECT CAT**. Verify the displayed frequency and mode against the radio.

Thetis / Hermes / ANAN: use the configured Thetis virtual serial CAT port and matching COM/baud settings. Defaults may not match your PC.

Yaesu FTDX10: select the FTDX10 profile, use the Enhanced COM Port, and match the radio CAT RATE. The adapter supports frequency/mode control only; do not assume FTDX10 PTT, CW keying, or digital transmit control.

Other displayed radio choices may be generic or planned profiles rather than completed native adapters. Verify compatibility before relying on them.

4. VAC audio and spectrum

Route receive audio from Thetis (or your source) to the Windows VAC recording/input endpoint. Select that endpoint in ARES-X and click REFRESH if necessary. Click **LISTEN TO INPUT** and confirm the audio level and spectrum respond.

If audio is absent, check Thetis VAC routing, Windows recording device selection, input level, mute state, and whether another program has exclusive control of the endpoint.

The spectrum is based on real mono VAC audio, not RF IQ. At 48,000 samples/second, the one-sided range is 0–24,000 Hz. The 16,384-point FFT improves bin spacing to about 2.93 Hz; it does not create a 192 kHz-wide RF spectrum.

5. WSJT-X FT4 / FT8 integration

WSJT-X is installed separately and remains responsible for actual FT4/FT8 modulation and decoding. ARES-X receives WSJT-X decoded/status messages through a local UDP bridge.

1. Start WSJT-X and configure its receive audio from the intended Thetis/VAC path.
2. Verify WSJT-X itself is decoding stations.
3. Open WSJT-X Settings → Reporting and enable UDP Server.
4. Use 127.0.0.1 and UDP port 2237 unless you configure a different matching port.
5. In ARES-X click CONNECT WSJT-X. Watch DECODER OUTPUT and EVENT LOG.

If no decodes appear in ARES-X, first verify WSJT-X decodes, then check UDP Server, matching host/port, firewall rules, and port conflicts.

6. Main workspace overview

Radio controls: profile, COM port, baud, CAT connection, frequency/mode readout, band selection, and VAC input.

Packet Format / Digital Mode: experimental AX-1 controls and local simulation tools. AX-1 is not a validated on-air digital mode.

Decoder Output: experimental decoder and bridged WSJT-X text.

Incoming Stations: incoming WSJT-X decode reports. Selecting a row is receive-only.

Event Log: connection and activity messages. In 0.9.16 it is above WSJT-X FUNCTIONS in the far-right column.

Spectrum / Waterfall: display of the selected VAC audio stream.

WSJT-X FUNCTIONS: bridge and selected-station actions.

7. Transmit safety

Selecting a station is receive-only. **REPLY TO SELECTED CQ / QRZ** sends a reply request to WSJT-X and may result in RF transmission if WSJT-X and the radio are configured to transmit.

Before any reply, verify callsign, frequency, mode, power, antenna/load, audio routing, PTT configuration, and licence privileges. Do not leave the station unattended.

HALT TX NOW sends a WSJT-X UDP Halt Tx request. Verify on the radio that transmission has actually stopped and the radio has returned to receive.

A tracker status or manual mark-complete action is not proof of a completed QSO or a WSJT-X log entry. Confirm and log contacts using your normal procedure.

8. Tools and X-TRACKER

X-TRACKER is the renamed Grid Tracker tab. It can use an explicitly supplied Maidenhead locator to plot a station. A callsign alone does not establish an exact location.

World Map is schematic unless a live-data source is explicitly shown. Satellite position/pass calculation is not guaranteed. Weather and HF propagation refreshes require internet and provider availability; propagation guidance is an estimate. Compass bearing is a manual visual aid, not a sensor.

9. Troubleshooting

CAT fails: check profile, COM port, baud, virtual port configuration, and whether another program owns the port. For FTDX10, use Enhanced COM and matching CAT RATE.

No VAC audio: select the recording/input endpoint rather than playback/output, refresh devices, and check Thetis routing and Windows recording level.

No WSJT-X decodes: confirm WSJT-X itself decodes, enable UDP Server, match host/port, click CONNECT WSJT-X, and check firewall/port conflicts.

Application fails to launch: run dist\\ARES-X\\ARES-X.exe. If the bundled runtime is damaged, rebuild from the extracted source folder using build_windows.bat. Keep the complete dist\\ARES-X folder together.

10. Known limitations and release notes

ARES-X is an experimental preview; no production readiness or full on-air validation is claimed. AX-1 is experimental. WSJT-X remains the external FT4/FT8 modem/decoder. The VAC spectrum is not an RF IQ spectrum. FTDX10 support is limited to frequency/mode CAT control. Some radio choices are placeholders/generic compatibility profiles.

Version 0.9.16 includes the Event Log relocation above WSJT-X FUNCTIONS, the X-TRACKER tab rename, and the 16,384-point VAC audio FFT in the main spectrum and Wide Graph.

No physical radio, antenna, or RF hardware test is certified by this package.

73 DE N2MDX
N2MDX UNLIMITED